HETEROGENEOUS FLEET VEHICLE ROUTING PROBLEM WITH TIME WINDOW: A CASE STUDY IN DKSH VIETNAM

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Abstract.

In this article, a company's healthcare business unit encountered a traffic problem that should be resolved by creating vehicle routes. According to the characteristics of the company, the problem is classified as a vehicle routing problem of a heterogeneous fleet with a time window, and the capacity and the number of vehicles is limited. Then a mathematical method is proposed and IBM CPLEX OPTIMIZER is used to solve the constraints. In order to evaluate the performance of the proposed method, experiments were conducted using real company data, and then compared with actual costs. It has been observed that the proposed method can be used to obtain reasonably better results for this problem. In addition, some ideas were put forward to actually implement the model and its effects in practice.

Keywords: Vehicle routing problem, Heterogeneous fleet of vehicle, Time window, Symmetric distance matrix.

1. Introduction

Nowadays, the efficiency of logistics activities is recognized as one of the factors that have the most profound influence on the overall performance of the business. The management of this logistics operation mainly includes managing and improving the process of inventory, import and export and transportation. Transportation problems with large volumes, high product diversity, limited vehicles, complex transportation networks, and time window constraints have always been a challenge for logistics managers. In search of a suitable answer to this challenge, more and more different approaches and variations of the vehicle routing problem (VRP) have been introduced by researchers in recent decades (Elshaer & Awad, 2020; Pan, Zhang, & Lim, 2021; Vidal, Laporte, & Matl, 2020). Among them can be mentioned mathematical optimization models that are developed increasingly complex with more and more real-world factors integrated. To continue this journey, as the main contribution of this article, a variant of the mathematical optimization model for heterogeneous fleet VRP was developed. Moreover, this proposed model is then applied to the actual operation of logistics enterprises in Vietnam as a validation process for later model improvements.

This paper is organized as follows: The next chapter introduces previous work regarding this topic. Chapter 3 presents methodology and chapter 4 is the results comparing to actual data, implementation and its impact. Finally, the last chapter offers conclusions and suggestions for future research.

2. Literature review

Hsu's research in 2007 aimed to determine the optimal transport routes within the constraints of time and number of vehicles for the perishable food distribution problem. This study uses stochastic mathematical models to solve vehicle routing problems with time window (VRP-TW). The results of the study show a significant share of inventory and energy costs in the system's cost structure. In addition, the authors also show the trade-off between inventory costs and fixed costs through adjustments for the number of vehicles. With stochastic time travel considerations, the results imply that time constraints become more difficult to satisfy than themselves under deterministic conditions (Hsu, Hung, & Li, 2007). In another study, published by Hande Öztop et al., mixed-integer programming and constraint programming models were introduced to identify optimal and near-optimal solutions for VRP-TW. In this study, heterogeneous vehicles are used to distribute valuable products such as cash to Automated Teller Machines. The resulting models provide safe routes with minimal travel time for heterogeneous vehicles. The limitation of the study mentioned by the authors is the small number of clients as a typical limitation of exact solution methods (Öztop, Kizilay, & Çil, 2020). In the global trend of reducing emissions, electric vehicles are also considered as a new mode of transport thereby creating another variant of VRP, electric VRP (EVRP). A study that developed an optimization model for EVRP with fuzzy uncertainty parameters was performed by Zhang et al. The above fuzzy uncertainty parameters describe the uncertainty conditions in EVRP such as energy consumption, travel time, service time, etc. In addition, the authors also propose an integrated algorithm with the core ALNS algorithm to enhance the solutions as well as optimize CPU time (Zhang, Chen, Zhang, & Zhuang, 2020). In research published in 2020, Hongqi Li considered the VRP-TW problem with modern unmanned aerial vehicles (UAVs). This study aims to optimize the delivery route of UAV-van vehicle complexes. The solution to the above problem was determined by the authors by the adaptive large neighborhood search algorithm. The results of this study are a reference for businesses about the feasibility of a new transportation mode (Li, Wang, Chen, & Bai, 2020). In 2020, Daniel proposed two different models for VRP-TW with limited time multiple trips. The numerical experiments of the study show that the proposed model is more efficient than the existing models mentioned by authors (Neira, Aguayo, De la Fuente, & Klapp, 2020).

Distribution networks are expected to become significantly different as distributors adopt customer picking through a network of locker facilities. A linear programming paradigm has been developed by Redi et al. to find a solution to VRP with such locker facilities. The numerical results of this study also demonstrate the advantages of customer picking strategy over home delivery strategy through the reduction of both important criteria, total transportation cost and number of vehicles (Redi et al., 2020). In Chen's research in 2020, a new optimization algorithm was developed for the problem of food distribution under COVID19-influenced conditions, which was developed on the basis of the integration of an artificial bee colony algorithm with tabu search. The performance of this proposed algorithm shows that artificial bee colony algorithms can be improved through the tabu search mechanisms. The numerical results also show that the number of vehicles will increase if distribution network operators want to increase the delivery speed. In addition, the maximum working time is extended if a higher delivery rate is required (Chen, Pan, Chen, & Liu, 2020). In 2021, a study evaluating VRP solutions that trade-off between the three pillars of sustainable development was conducted by Hassana Abdullahi et al. The solutions in this study are determined through a metaheuristics algorithm that reconciles the advantages of a greedy local search method and a biased randomized savings heuristic. The three sustainability pillars are integrated in this study in the form of objective functions, which are costs representing economic factors, accident risks representing social factors, and CO2 emissions representing environment factors (Abdullahi, Reyes-Rubiano, Ouelhadj, Faulin, & Juan, 2021).

3. Methodology

First, the problem and its objective are defined. Then, research papers related to the problem are searched and review. Next, based on research papers and reality, a model to solve the problem is construct and data is collected and analyzed. After that, by using the analyzed data as input, the model will be run and created a result. The result is then validated by comparing with the reality. If the outcome is not better than reality, the model will go back to problem modelling, data analyzing and running model and result validating again until it is better than reality. Finally, the conclusion and suggestion are provided based on the better outcome.

This paper proposes a mathematical model to solve a real-life heterogeneous fleet vehicle routing problem with time windows. The problem consists in determining, each day, how to allocate the trucks from depots to all customers, the amount to be delivered in each truck to each customer, which one is the best route and the suitable time for attending each node, with the aim of minimizing the total transportation cost (summation of fixed and operating costs), satisfying the customers demand and respecting all the problem constraints (vehicle number and feature, and time windows). The difference of this model comparing to the papers is, in this paper, a linearization method is introduced to transform two non-linear constraints into linear constraints. It also considered different types of products with different space, which both papers did not take into account. A directed graph G = (N, A) is given, where $N = \{1, 2, \ldots, n\}$ is the set of n nodes. Node 1 represents the depot while the remaining node set $N \backslash \{1\}$ corresponds to the customers from node 2 to node n. Node set N has 3 indexes: i, j, $k \in N$. Each pair of location (i,j), is associated with distance d_{ij}. There are $p \in P$ types of products, defined by their occupied space $Spacep$ in a vehicle. Node $i \in N \backslash \{1\}$ (all customers except the depot) have demand Dip for product type

$p \in P$. M is the set of transportation modes, each vehicle $m \in M$ has their capacity $Capm$, fixed cost Fm, total operating cost $cmdijm$, velocity vm and service time sim at node $i \in N \backslash \{1\}$. Whenever a vehicle is triggered, it will use a fixed amount of money for maintenance and capital amortization costs, no matter how far it goes. This is called fixed cost Fm. Operating cost is the sum of fuel and oil on average consumed by a vehicle. The total operating cost of a vehicle on route (i, j) equaled to operating cost per kilometer traveled cm multiplied with the distanced travelled on route (i, j) by vehicle m $dijm$. The time window for all nodes $i \in N \backslash \{1\}$ are [e, l] with e is the earliest time and l is the latest time vehicles can enter that node. Time window for depot is [0, L], all trucks can start leaving the depot at 0 and must be back by L. The parameters and decision variables of the proposed model are presented in Tables 1 and 2.

Table 1. Model's parameters

Parameter	Description
D_{ip}	The demand for customer node i of product type p
$Space_p$	The space of product type p
d_{ij}	Distance between node i and node j
F_m	Fixed cost when using vehicle type m
C_m	Operating cost per kilometer travelled of vehicle type m
Cap_m	Capacity of vehicle type m
v_m	Velocity of vehicle type m
S_{im}	Serving time at customer node i of vehicle m
e	Earliest time to arrive at customer zone
l	Latest time to arrive at customer zone
L	Latest returning time of all vehicles to the depot

Table 2. Model's decision variables

Parameter	Description
x_{ij}^m	Equal to 1 if vehicle type m travels directly from node i to node j and equal to 0 if otherwise
y_{ijmp}	Quantity of product type p that vehicle m carries when it leaves node i to service node j
a_{im}	Arriving time of vehicle type m at node i
l_{im}	Leaving time of vehicle type m at node i

As shown in equation (1), the model's objective function focuses on minimizing total transportation cost, including a fixed cost (F_m) and operating cost $(C_m d_{ij})$. Each cost is then multiplied to variable x_{ij}^m to make sure that only cost of triggered vehicle will be summarized. The decision variable x_{1j}^m means that only vehicle started from the depot (node $i = 1$) by vehicle m to any customer (node j) will be counted with fixed cost since this is where all vehicles start.

$$\boldsymbol{Minimize} \sum_{m \in M} \sum_{j \in N} F_m \, x_{1j}^m + \sum_{m \in M} \sum_{\substack{i,j \in N \\ i \neq j}} C_m d_{ij} \, x_{ij}^m \tag{1}$$

$$\sum_{m \in M} \sum_{i \in N} x_{ij}^m \ , \forall j \in N \backslash \{1\} \tag{2}$$

$$\sum_{m \in M} \sum_{j \in N} x_{ij}^m \ , \forall i \in N \backslash \{1\} \tag{3}$$

$$\sum_{\substack{i \in N \\ i \neq j}} x_{ij}^m = \sum_{\substack{k \in N \\ k \neq j}} x_{jk}^m \, , \forall m \in M, j \in N \backslash \{1\} \tag{4}$$

Constraints (2), (3) and (4) makes sure that each customer is visited only once and that if a vehicle visits a customer, it must also depart from it. Each vehicle can be used no more than once is imposed by constraints (5) and (6).

$$\sum_{j \in N \backslash \{1\}} x_{1j}^m \leq 1, \ \forall m \in M, i \in N \tag{5}$$

$$\sum_{i \in N \backslash \{1\}} x_{i1}^m \leq 1, \ \forall m \in M, j \in N \tag{6}$$

Constraint (7) makes sure that there will be no vehicle assigned to route (i, j) when node i and j are the same.

$$\sum_{m \in M} x_{ij}^m = 0, \ \forall (i, j) \in N, i = j \tag{7}$$

Constraint (8) is the commodity flow constraints: they specify that the difference between the quantity of goods a vehicle carries before and after visiting a customer is equal to the demand of that customer and constraints (9) ensure that the vehicle capacity is never exceeded.

$$\sum_{m \in M} \sum_{\substack{i \in N \\ i \neq j}} y_{ijmp} - \sum_{m \in M} \sum_{\substack{k \in N \\ k \neq j}} y_{jkmp} = D_{ip}, \forall p \in P, j \in N \backslash \{1\} \tag{8}$$

$$\sum_{p \in P} D_{ip} \, Space_p x_{ij}^m \leq \sum_{p \in P} y_{ijmp} Space_p \leq \left(Cap_m - \sum_{p \in P} D_{ip} Space_p\right) x_{ij}^m, \forall (i, j) \in N, i \neq j, m \in M \tag{9}$$

Constraint (9) prevent any product is taken back to the depot while constraint (10) guarantees that no products will be assigned if a vehicle is not activated.

$$y_{i1mp} = 0, \forall i \in N, m \in M, p \in P \qquad (10)$$

$$\sum_{p \in P} y_{ijmp} \leq x_{ij}^m \times BigM, \forall (i,j) \in N, m \in M \qquad (11)$$

Constraints (12) make sure that if customer j follows customer i in the route, the arrival time at customer j is equal to the departure time from customer i plus the travel time between these two customers.

$$\left(a_{jm} - \left(l_{im} + \frac{d_{ij}}{v_m}\right)\right) x_{ij}^m = 0, \forall (i,j) \in N, i \neq j, m \in M \qquad (12)$$

Constraints (13) relate arrival time, departure time with service time and guarantee that their relationships are compatible to the time window. Constraints (14) and (15) are the time window for the customers and depot. Finally, constraints (16), (17), (18) and (19) restrict the values of the variables.

$$\left(l_{im} - (a_{im} + s_{im})\right) x_{ij}^m = 0, \forall i \in N \backslash \{1\}, j \in N, i \neq j, m \in M \qquad (13)$$

$$a_{jm} \geq e \sum_{j \in N} x_{ij}^m, \ \forall i \in N \backslash \{1\}, m \in M \qquad (14)$$

$$a_{jm} \leq l \sum_{j \in N} x_{ij}^m, \ \forall i \in N \backslash \{1\}, m \in M \qquad (15)$$

$$x_{ij}^m \in \{0,1\}, \ \forall (i,j) \in N, m \in M \qquad (16)$$

$$y_{ijmp} \geq 0, \ \forall (i,j) \in N, m \in M, p \in P \qquad (17)$$

$$a_{im} \geq 0, \ \forall i \in N, m \in M \qquad (18)$$

$$l_{im} \geq 0, \ \forall i \in N, m \in M \qquad (19)$$

In order to linearization the non-linear constraints (12) and (13), the BigM constraints method was applied and formulation the alternative constraints (20), (21), (22) and (23) below.

$$a_{jm} \geq \ l_{im} + \frac{d_{ij}}{v_m} - \left(1 - x_{ij}^m\right) BigM, \forall m \in M, (i,j) \in N, i \neq j \qquad (20)$$

$$a_{jm} \leq \ l_{im} + \frac{d_{ij}}{v_m} + \left(1 - x_{ij}^m\right) BigM, \forall m \in M, (i,j) \in N, i \neq j \qquad (21)$$

$$a_{im} \geq \ l_{im} - s_{im} - \left(1 - x_{ij}^m\right) BigM, \forall m \in M, i \in N \backslash \{1\}, j \in N, i \neq j \qquad (22)$$

$$a_{im} \leq \ l_{im} - s_{im} + \left(1 - x_{ij}^m\right) BigM, \forall m \in M, i \in N \backslash \{1\}, j \in N, i \neq j \qquad (23)$$

4. Case study and numerical results

DKSH Vietnam Co., Ltd. are the leading Market Expansion Services provider for companies who want to grow their business in Vietnam. Serving their business partners through their extensive global networks and industry expertise, DKSH Vietnam help companies to grow their businesses in new and existing markets. As one of Vietnam's leading business organizations, their network of 22 business locations, including offices, distribution centers, and cross-docks, across the country and 4,500 employees make them an international "local" company. DKSH Vietnam is committed to building long-term business partnerships in Vietnam. DKSH Vietnam constantly strives to bring world-class standards to industries in Vietnam while also contributing to the development of local communities.

In Healthcare branch, DKSH Vietnam has three warehouses located in Ha Noi, Da Nang and Binh Duong, and a distribution center (DC) in Can Tho. All oversea products are shipped to Binh Duong warehouse from Cat Lai seaport. Then, products will be transported to Ha Noi and Da Nang warehouses depending on the customers' location. The warehouse main functions are received products, pick and pack according to customers order, release bills and ship to customers. The Can Tho DC main responsibility is only to receive packages in large containers from Binh Duong warehouse and distribute packages into smaller truck and ship to customers. The Da Nang warehouse covers from Quang Binh to Binh Thuan. From Quang Binh up to the north is covered by Ha Noi warehouse, and from Binh Thuan province down to the south is cover by Binh Duong warehouse.

Right now, DKSH Vietnam delivery policy is that one truck is responsible for one province's order. The capacity of the truck is depended on the province's demand. Trucks will get packages from warehouses and transport to the province's city. From there, a local transportation team hired by DKSH Vietnam consisting of 3-5 people, depending on the province size, will receive packages from the truck and shipped to customers. After distributing packages, the truck will ship to customers living in the city only and come back to warehouses after finish shipping. The latest time for shipping is 6 PM then all trucks must go back and be available at the depot before 1 AM of the next day. Customer orders cut off time is 6 PM each day. Orders are then sent to warehouses, where they will be processed, pick and pack into packages, and loaded on trucks. Only after the cut off time will DKSH Vietnam know exactly each customer demand for the next delivery and each truck delivery schedule will be generated. By 2 AM the next morning, trucks are loaded with packages and the transportation cycle begins again. Each province or city is divided into many areas with different delivery schedule. Delivery frequency ranges from 1 to 6 days a week, depending on customer need and distance from the warehouse. Usually, big city centers will be delivered 6 days a week. DKSH Vietnam continues to expand its market by signing contracts with new clients each year. In 2018, Sanofi became DKSH Vietnam’s new client and is one of the biggest clients. Its sales last year accounted for 28% of total sales. DKSH Vietnam classified their customers into 4 groups as Table 3.

Table 3. DKSH Vietnam customer classification

Group	Average value per order in a month (VND)	Proportion
A	Over 15 million	5%
B	10 – 15 million	12%
C	5 – 10 million	28%
D	Under 5 million	55%

As can be seen, more than half of customers usually place orders under 5 million VND and the average SKU per order is 5. This means that each time customers place an order they usually buy few products and spend little money on it.

4.1. Case study data

Customers demand data collected from DKSH Vietnam is processed by grouping customer in some districts in a province into one node. As can be seen, some nodes are different areas of one province. Unlike DKSH Vietnam transportation rule, where they decided that one vehicle is responsible for one province, in this model, the vehicle can transport freely between nodes that are not in one province. One month demands in 2018 of these nodes were chosen randomly and the average is used for the model customers demand input as Table 4.

Table 4. Node demand

Node	SKU				
	1	2	3	4	5
1	0	0	0	0	0
2	14	8	11	5	2
3	10	13	5	9	1
4	5	6	3	2	2
5	3	2	7	5	0
6	6	5	2	5	4
7	9	11	5	7	2
8	3	5	5	2	0
9	5	2	2	3	1
10	6	5	2	2	1
11	7	4	4	1	2
12	7	3	2	1	1
13	6	4	2	1	0
14	4	3	5	1	0
15	6	2	0	1	3
16	7	1	5	2	4
17	9	3	0	2	0
18	12	2	0	0	2
19	5	7	2	2	1
20	4	1	0	5	4
21	3	1	1	2	1
22	14	2	0	3	0
23	6	2	1	4	0
24	4	6	3	10	4
25	7	5	2	9	2
26	5	7	9	0	3
27	11	9	5	3	2
28	2	1	3	0	5
29	9	1	1	2	0
30	11	2	3	7	4
31	14	2	1	0	0
32	9	2	3	1	0
33	8	3	2	3	2
34	12	4	3	0	0
35	14	3	2	0	1
36	5	2	5	3	0
37	10	3	4	0	0
38	9	4	0	5	1
39	7	3	2	1	0
40	7	3	0	3	0
41	8	2	1	0	1
42	6	4	0	3	0
43	5	2	3	1	0
44	9	2	3	0	0
45	7	5	1	2	1

Each node demand contains many packages of different size, each package is for a customer as Table 5. The distance between these nodes is estimated based on the position where the truck drops off packages to a local transportation team and are calculated by using Python linked with Google map.

Table 5. Product space requirement

SKUs	1	2	3	4	5
Space (m^3)	0.1	0.2	0.3	0.4	0.5

All vehicles will start leaving the depot at 2 AM each day. This is considered as time 0 in the model when all transportation begins. Each vehicle type has different capacity, velocity. To create equality between vehicles with the same type, their parameters (capacity, velocity, operating cost per km travelled, service time for each customer will be exactly the same. This means, all trucks of the same type will have equal probability chosen by the model. Vehicle velocity may varies depended on the size of the road. So, their average velocity will be taken, which is 40 – 45km/h, depending on the truck size. Bigger truck will have slower speed. These data are consolidated by DKSH Vietnam given data as well as reference by sShip transportation company trucking fee as Table 6. Meanwhile the time window for customer ($[e, l]$) is 06:00-14:00 with the latest depot depart time (L) is 01:00 every day.

Table 6. Vehicle characteristics

Vehicle ID	1-10	11-20	21-23
Vehicle type (ton)	2	2.5	15
Capacity (m3)	13	16	66
Velocity (km/h)	45	45	40
Fixed cost (VND)	210,000	230,000	350,000
Operating cost (VND/km)	21,500	22,500	31,500
Service time (hour)	0.3	0.4	0.5

4.2. Optimization results

In Table 7, the optimization results are presented, which proposed the total transportation cost is 611,174,000 VND (~27,407 USD) with 16 vehicles are triggered.

Table 7. Vehicle routing optimization result

Vehicle ID	Route Node Sequence
1	Depot-2-19-Depot
2	Depot-4-5-6-Depot
3	Depot-25-39-43-45-Depot
4	Depot-37-38-Depot
5	Depot-21-44-Depot
6	Depot-11-10-Depot
7	Depot-7-27-Depot
8	Depot-29-26-28-30-Depot
9	Depot-16-Depot
10	Depot-40-Depot
11	Depot-34-3-Depot
12	Depot-24-35-20-Depot
14	Depot-15-14-32-31-Depot
15	Depot-22-23- Depot
18	Depot-33-12-13-42-Depot
20	Depot-8-9-36-17-18-41-Depot

These results are compared with the cost of routes actually covered by the trucks of the company at the same time as Figure 1. In both cases, the route cost is calculated by adding the fixed and operating costs.

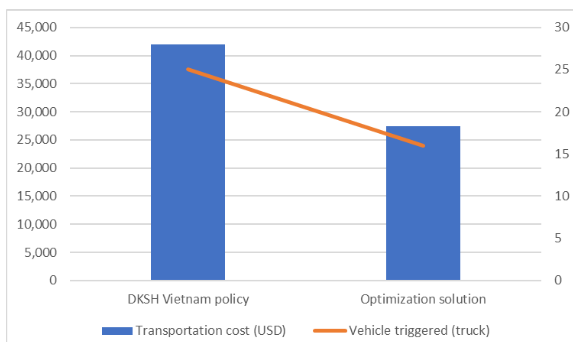

Figure 1. Transportation performance

As state before, there are 45 nodes comes from 25 provinces. With DKSH Vietnam transportation rule, which is one vehicle responsible for one province, there are 25 triggered vehicles and it costs them around 42,000 USD transportation cost. This does not even cover the transportation cost when the trucks go around the province's city for delivery. Comparing both models, it is clear that with the new model, DKSH Vietnam can reduces 9 trucks, saving around 14,500USD per day, or 3,253,480USD per year.

The advantage of this proposal is that each vehicle does not strictly need to be visited only one province. By dividing provinces into one or many nodes based on their demands, a vehicle can flexibly travel between two nodes of, maybe, different provinces, as long as this route saves more cost. Another advantage of this is that instead of using a big truck to deliver to customers in cities, the truck will now just drop off packages and one or two local transportation employees, depend on how many customer orders, will deliver them to customers. By switching from a truck to motorbikes, packages will be delivered faster, considering Vietnam traffic characteristics. Furthermore, it will save more transportation cost since a 2-ton truck cost 21,500VND/km while two motorbikes cost only 10,000VND/km (5,000VND/km each). For a more practical and out of scope comparison, transportation employee costs will also now be included. There are two types of transportation employee, local transportation employee, and on-truck transportation employee. Local transportation employees are DKSH Vietnam employees who based in each province with their duty to receive a package from a truck and distribute it to customers. On-truck transportation employees are the ones who go with the truck to help the driver distributing packages to local employees and deliver packages to the city’s customers while the driver stays with the truck. With the new method, the on-truck employees are not needed anymore, since now all trucks do not deliver directly to any customer. In this particular situation, 25 of on-truck employees are a shift to 25-35 local employees. Salaries for these two types of employees are 6 million VND/person/month. With about 10 people differences, it will cost around 60 million VND/month more (~2,690USD). However, this is not much comparing to the 14,500USD/day saving in transportation cost.

5. Conclusion

This paper proposed a mathematical model to solve a real-life transportation problem in a pharmaceutical distribution company, DKSH Vietnam. The problem is classified as a heterogeneous fleet vehicle routing problem with time windows problem, which was modified to a real-life situation and introduced a linearization method for some constraints.

The proposed approach has fulfilled the research objective, which is to minimize total transportation cost by determining the optimal set of routes for a fleet of vehicles. It has offered a better solution in terms of distribution cost and a number of trucks used compared to the current practice while at the same time still respect all problem constraints and time window. DKSH Vietnam could apply this model to reality easily, save more transportation and system cost than what they are doing right now.

Most of the problem aspects have been covered in this model, however, there are still some limitations due to solving time constraint. For future work, the solving time should be reduced to at most 30 minutes to reach optimal solution. In order to do that, other solving method, like heuristics, could be developed. The model could be improved to be applied for more complex problems at Southern region, where a DC is involved, and Middle region since there are longer distance and the lead time varies from 1 to 2 days. For a more practical and complete solution, other transportation factors such as toll, rush hour, reducing emission gas should also be taken into consideration. The number of product types could increase more to be more realistic.

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